SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1513

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I **K.Sasidhar Reddy(192525140)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: K.Sasidhar Reddy

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
1. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.
2. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
3. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
4. A development team wants to adopt containerization for application deployment.
   Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.

Provide an example of a microservices-based application.

1. Handling Seasonal Traffic Spikes

> **Scenario:** *An online retailer, similar to Amazon or Flipkart, runs Big Billion Days / Black Friday sales where traffic can jump 10x to 50x within minutes of a flash deal going live.*

How auto-scaling and load balancing work together

Auto-scaling automatically adds or removes compute instances (VMs or containers) based on demand. The company defines a minimum and maximum fleet size and scaling policies, for example "add two instances when average CPU exceeds 65% for two minutes". Scheduled scaling can also pre-warm capacity before a known sale starts. A load balancer sits in front of the fleet and distributes incoming requests across all healthy instances, using algorithms such as round robin or least connections. When auto-scaling launches a new instance, it registers with the load balancer and starts receiving traffic; when demand falls, instances are drained and terminated.

- **Elasticity:** Capacity follows demand, so the site is never under-provisioned during the spike or over-provisioned afterwards.
- **Even distribution:** No single server is overwhelmed, avoiding hotspots and slow responses.
- **Health checks:** The load balancer stops sending traffic to failed instances, and auto-scaling replaces them automatically.
- **Multi-zone deployment:** Instances are spread across several Availability Zones, so one data-centre failure does not take the store offline.

Reducing downtime and improving user experience

Because failed or overloaded servers are removed and replaced automatically, there is no single point of failure and no manual intervention in the middle of a sale. Shoppers see consistently fast page loads, search results and checkout, rather than timeouts, "503 Service Unavailable" errors or lost carts. Faster, reliable checkout directly raises conversion rates and customer trust, and avoids revenue loss during the most valuable hours of the year.

Role of monitoring tools

Monitoring tools (Amazon CloudWatch, Azure Monitor, Google Cloud Monitoring, Datadog, Prometheus/Grafana) provide the metrics that drive the whole system.

- **Trigger scaling:** CPU, memory, request count per target and response latency feed the auto-scaling policies.
- **Alerting:** Alarms notify engineers of rising error rates, slow database queries or failing health checks before customers are affected.

- **Dashboards and logs:** Real-time views of orders per minute, cart abandonment and latency help the team make quick decisions.
- **Capacity planning:** Post-sale analysis of metrics shows how to tune thresholds and budgets for the next event.

Real-world flow

At 8:00 PM a flash sale begins. Traffic rises sharply, and CloudWatch detects average CPU above the threshold. Auto-scaling grows the web tier from 10 to 120 instances within minutes. The Application Load Balancer spreads requests across all zones, and a failing instance is detected and replaced without any shopper noticing. After the sale ends, the fleet scales back to 10 instances, so the company pays only for what it used.

2. Secure and Cost-Effective Backup Storage

> **Scenario:** *An organization needs to keep many terabytes of backups (databases, file servers, logs) for years, securely and affordably.*

Durability and data redundancy

Cloud object storage (Amazon S3, Azure Blob Storage, Google Cloud Storage) is designed for extremely high durability, commonly quoted as 99.999999999% ("eleven nines"). This is achieved through redundancy:

- **Replication and erasure coding:** Each object is broken into pieces and stored across multiple disks and devices, so failure of individual hardware does not lose data.
- **Multi-zone storage:** Data is automatically copied across several Availability Zones within a region.
- **Cross-region replication:** Copies can be kept in a geographically distant region to survive regional disasters.
- **Self-healing and integrity checks:** The service continuously verifies checksums and rebuilds corrupted or missing fragments.
- **Versioning and object lock:** Previous versions are retained, and write-once-read-many (WORM) settings protect against accidental deletion and ransomware.

Protecting sensitive backups: encryption and access control

- **Encryption in transit:** TLS secures data as it travels to the cloud.
- **Encryption at rest:** AES-256 encryption is applied to stored data, using provider-managed keys or customer-managed keys in a Key Management Service (KMS/HSM). Customer-held keys give extra control and allow keys to be revoked.
- **Identity and Access Management (IAM):** Least-privilege policies and role-based access ensure only authorized users and services can read or restore backups.
- **Multi-factor authentication and MFA-delete:** Extra verification is needed for sensitive actions such as deleting backups.
- **Network controls and auditing:** Private endpoints, bucket policies and access logs restrict and record who touched the data.

Cost advantages over traditional storage

Aspect	Traditional on-premises	Cloud storage
Upfront cost	Large capital expense for arrays, tape libraries and data-centre space	No upfront purchase; pay-as-you-go operating expense
Capacity planning	Must buy for peak future needs; often over-provisioned	Scales on demand; pay only for what is stored
Maintenance	Hardware refresh, power, cooling, staff	Handled by the provider
Tiering	Manual and limited	Automated lifecycle rules move old data to low-cost archive tiers (e.g. S3 Glacier)
Off-site copy	Requires a second site or tape transport	Built-in geographic replication

Disaster recovery example

A hospital group stores nightly database backups in cloud storage with cross-region replication. A ransomware attack encrypts its primary servers and local backups. Because the cloud copies are versioned and protected by object lock, the IT team restores the last clean version, spins up virtual machines in the secondary region and brings the patient-record system back online within hours. Recovery Point Objective (RPO) is at most one day of data, and Recovery Time Objective (RTO) is hours instead of days, with no purchase of a standby data centre.

3. Global Deployment with CDNs

> **Scenario:** *A mobile app company launches a real-time multiplayer game and live video streaming feature for users in Asia, Europe and the Americas.*

How CDNs improve performance and reduce latency

A Content Delivery Network is a globally distributed network of edge servers (Points of Presence) that cache and serve content from locations close to the user. Instead of every request travelling to a single origin server on another continent, the request is answered by a nearby edge node. This shortens the physical distance, reduces the number of network hops, offloads traffic from the origin and speeds up loading of static assets (images, app updates, video segments, scripts).

How edge servers enhance user experience

- **Lower latency:** A player in Chennai connects to a nearby edge node rather than a server in Virginia, cutting round-trip time from hundreds of milliseconds to tens.

- **Faster start-up and less buffering:** Video segments are cached at the edge and delivered with adaptive bitrate streaming.
- **Traffic absorption:** Edge nodes handle huge audiences (for example a live tournament) without overloading the origin.
- **Edge computing:** Authentication checks, request routing, image resizing and DDoS filtering can run at the edge.
- **Optimized connections:** Persistent connections, TLS termination and modern protocols (HTTP/3, QUIC) improve speed on unstable mobile networks.

How cloud providers support global availability

- **Global regions and Availability Zones:** Applications are deployed in multiple regions close to users.
- **Global DNS and traffic routing:** Services such as Route 53 or Traffic Manager use latency-based or geo-based routing and automatic failover.
- **Multi-region databases:** Globally replicated databases (e.g. DynamoDB Global Tables, Cosmos DB, Spanner) keep data close to users.
- **Global backbone networks:** Private fibre between regions is faster and more reliable than the public internet.
- **Redundancy and failover:** If one region fails, traffic is shifted to another.

Streaming and gaming example

For a live streaming platform, the video is ingested once, transcoded into several bitrates and pushed through a CDN, so millions of viewers pull segments from nearby edges. For a multiplayer game, static assets and patches come from the CDN, while game sessions run on regional servers chosen by latency-based routing. Matchmaking places players on the nearest region, keeping ping low, which is vital because even 100 ms of extra delay is noticeable in fast-paced games. If a region becomes unavailable, players are reconnected to the next closest one.

4. Compliance and Security for a Financial Institution

> **Scenario:** *A bank moves its online payments platform to the cloud and must satisfy regulators and auditors.*

Regulatory compliance and auditing support

Cloud providers operate under a shared responsibility model: the provider secures the underlying infrastructure (facilities, hardware, hypervisor), while the customer secures its own configuration, data and access. Providers undergo independent audits and publish certifications and attestations such as PCI DSS, ISO 27001, SOC 1/2/3, and GDPR support, as well as region-specific frameworks. They make these reports available (e.g. AWS Artifact, Azure Compliance Manager) so the bank can demonstrate compliance to regulators such as the RBI in India, and offer compliance tools that continuously check configurations against standards (AWS Config, Azure Policy, Security Command Center).

Role of logging, monitoring and encryption

- **Logging:** Services such as CloudTrail, Azure Activity Log and Cloud Audit Logs record every API call and administrative action (who, what, when, from where). Logs are stored immutably and retained for audit and forensic investigation.
- **Monitoring and threat detection:** Tools such as SIEM platforms, GuardDuty and Defender for Cloud analyse logs and network activity in real time, detect anomalies (unusual logins, large data transfers) and trigger automated alerts or responses.
- **Encryption:** Data is encrypted in transit (TLS) and at rest (AES-256), with keys managed in HSM-backed key services. The bank can use customer-managed keys, rotate them regularly and revoke access instantly.
- **Identity controls:** MFA, least-privilege IAM and segregation of duties limit insider and external threats.

Ensuring data sovereignty

- **Region selection:** Deploy and store data only in approved regions or countries (for example Indian regions for data that must remain in India).
- **Policy guardrails:** Use organization policies or service control policies to block creation of resources in non-approved regions.
- **Controlled replication:** Configure backups and disaster recovery to replicate only within permitted geographies.
- **Key control:** Keep encryption keys in-country, using customer-managed keys or external key stores, so the provider cannot access plaintext data.
- **Contracts and sovereign options:** Use contractual data-processing terms, provider commitments on data residency and, where needed, sovereign or dedicated cloud offerings.

Example: sensitive financial transaction

A customer makes a card payment. The request is encrypted with TLS, and the card number is tokenized so the real value is never stored in application databases. The transaction record is written to an encrypted database in an in-country region, and the encryption key is held in an HSM-backed KMS under the bank's control. Every access to the record is logged by the audit service; a monitoring system flags an unusual pattern, such as many high-value transfers from a new device, and automatically blocks the account and alerts the fraud team. During an audit, the bank produces the immutable logs and the provider's PCI DSS attestation as evidence of compliance.

5. Containerization for Application Deployment

> **Scenario:** *A development team wants to package and deploy an application consistently across development, testing and production.*

Containers vs. virtual machines

Aspect	Virtual machines	Containers
Architecture	Each VM runs a full guest operating system on a hypervisor	Containers share the host OS kernel and isolate processes
Size	Gigabytes	Megabytes
Start-up time	Minutes	Seconds or less
Resource usage	High overhead: each VM needs its own OS, CPU and memory	Lightweight; many more containers fit on the same hardware
Performance	Slight overhead from virtualization layer	Near-native performance
Isolation	Strong, hardware-level isolation	Process-level isolation; slightly weaker, but improved by sandboxing
Best for	Different operating systems, strong isolation, legacy apps	Microservices, CI/CD, rapid scaling

How orchestration (Kubernetes) simplifies deployment and scaling

Running a few containers by hand is simple, but running hundreds across many machines is not. Kubernetes automates this:

- **Automated scheduling:** Places containers on suitable nodes based on resource needs.
- **Horizontal scaling:** Increases or decreases the number of replicas automatically (Horizontal Pod Autoscaler) according to load.
- **Self-healing:** Restarts failed containers, replaces unhealthy nodes and reschedules workloads.
- **Rolling updates and rollbacks:** Deploys new versions gradually with no downtime and reverts quickly if something fails.
- **Service discovery and load balancing:** Gives services stable names and distributes traffic between replicas.
- **Declarative configuration:** Desired state is described in YAML files, and Kubernetes continually works to match it, which fits well with CI/CD and infrastructure as code.
- **Config and secrets management:** Separates settings and credentials from the container image.

Portability across cloud environments

A container image bundles the application with its libraries and dependencies, so it behaves the same on a developer laptop, an on-premises server or any cloud. This removes the "it works on my machine" problem. Because Kubernetes is an open standard offered as managed services

(Amazon EKS, Azure AKS, Google GKE) and can also run on-premises, teams can move workloads between providers, run hybrid or multi-cloud setups and avoid vendor lock-in. Container registries and open standards (OCI) further simplify distribution.

Example: microservices-based e-commerce application

An online store is split into independent services, each in its own container:

- **Frontend / API gateway:** Serves the web and mobile apps and routes requests.
- **Product catalogue service:** Manages product data and search.
- **Cart service:** Maintains shopping carts, backed by an in-memory store such as Redis.
- **Order service:** Creates and tracks orders.
- **Payment service:** Integrates with payment gateways.
- **User/authentication service:** Handles accounts and login.
- **Notification service:** Sends emails and SMS messages.

Each service has its own codebase, database and release cycle. During a sale, Kubernetes scales only the cart and order services, while the notification service stays small. If the payment service crashes, Kubernetes restarts it without affecting browsing. A team can update the catalogue service with a rolling deployment without redeploying the whole application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity